

How SoMs Take The Effort Out of Designing

The system-on-module (SoM) approach allows the fastest time to market. Utilising off-the-shelf standard components like a baseboard and SoM, designers are able to start developing applications even before the custom baseboard is complete



DM816x/C6A816x/AM389x evaluation module for evaluating DM816x DaVinci processors, C6A816x C6-Integra DSP+ARM processors and AM389x Sitara ARM MPUs

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"System-on-module is an extension of the concept of system-on-chip which lies between a full-up computer and a micro-controller in nature"

— Texas Instruments

“We wanted to assist product designers in developing products with reduced time, effort and cost, to foster innovation. This was our aim when we conceptualised the SoM based on TI's DaVinci DM816x and DM814x video processors,” says Dhaval Parikh, hardware division head at Elnfochips.

But what exactly is an SoM? A system-on-module (SoM), also known as computer-on-module (CoM), is a complete embedded computer system built on a single circuit board. It includes the microprocessor, input/output (I/O) peripherals, memory,

power management system and other components on the same chip.

How is a SoM different from a SoC?

SoM is the combination of a system-on-chip (SoC) with memory (DDR2, DDR3 or Flash), power management and other components embedded into one module. Moreover, SoMs do not need processor 'wake up' as in SoCs. 'Wake up' is the process of getting a shut-down or sleeping CPU back to work and fully operational.

Since the SoM usually lacks standard connectors for I/O peripherals on the board, it is mounted on a carrier board or baseboard. The baseboard has standard peripheral connectors for Ethernet, USB, etc. The wiring of these connectors from the baseboard to the SoM is done via buses. By separating the design of the SoM and baseboard, modular design concepts can be imple-

mented in the system.

In contrast, SoCs are typically I/O-specific devices. So SoCs from even the same manufacturer may not be interchangeable.

Why modular?

Debugging complicated silicon hardware and related hardware for a special application usually requires a high design overhead. By using a module-based approach, it is much easier to upgrade a CPU component. You need to change only the module to be upgraded, leaving the baseboard essentially intact. This reduces the overhead incurred during development and allows designers to spend more time on applications that are key to product success. This leads to reduced business risk and a lower design overhead. Also, one has multiple CPU choices with a modular approach. The best part is hardware and software can be developed in parallel as developers can use off-the-shelf hardware with the same processor core as their product.

Another important point is that the modular approach reduces the risk of having an end-of-life product due to memory, chipset or CPU going kaput. With the capability to swap in another CPU SoM board, the system can be kept running with minimal changes to the baseboard. This also allows the user to enjoy scalability with multiple-speed cores for similar products, which, in turn, allows high-volume pricing to bring down the cost. A lot of applications in the embedded space are shipped in relatively low volumes, which work very well with the concept of a plug-in CoM.